

Satellite Navigation

EMLID REACH RS2

Setup and annotations for a measurement campaign

Group: group 1

1 Configuration

The Reach receivers interface can be accessed via browser (or software) or the Emlid Flow App (smartphone). The app has some additional features, installation is not mandatory however.

1. **Connecting:** Turn on the Reach receiver

- Smartphone/App: activate Bluetooth, open app and *refresh* until the device shows up. Select and connect.
- Browser: Reach and device must be in the same network.
 - Connect using the IP address and password of the network
 - Use the Reach's own network (recommended)
IP: 192.168.42.1
PW: emlidreach

Contact your exercise supervisor for questions or assistance.

2. **Status:** Here you can see details about the current reception, location and some settings you have made.

3. **Wi-Fi** → Hotspot mode → Activate, connect to network

4. **Base Settings** → Configure → Measure and set antenna height: 1.534m

5. **Logging** Log and Settings → Gear icon:

- Raw data: Rinex 3.04, Default
- Info for RINEX header: *optional*
- Position track: On, NMEA
- Base correction: Off
- Full memory behaviour: Stop when memory is full
- Split log intervall: 24h
- Auto start, Backup source, Raw data debug: Off

6. **Settings** → GNSS settings

- Elevation mask: 5 degree
- SNR mask: 10
- GNSS-constellations: GAL, GPS, BeiDou
- Logging rate: 10 Hz

7. **Correction input** → Off

8. **Base output** → Off

Note: Bottom bar menu: Survey → + → Create new project: Map view with current location

2 Recording and Saving

- **Recording:** Logging → starten (speichert auf Reach)
- **Daten kopieren:** Loggen → Protokolle und Einstellungen → Herunterladen
- Hinweis: Kontrollieren Sie die Daten auf Vollständigkeit und Inhalt.

3 Measurement Campaign

Notes and remarks for the measurements.

- **Lab test:** Test your setup in the lab to verify the devices are working as intended..
- **Static measurement:** Think about suitable conditions for a static measurement. Write start and end time of each measurement. Alternatively, you can divide the measurement into different files.
- **Kinematic measurement:** Think about suitable conditions for a kinematic measurement. Are the settings still correctly chosen for the new scenario? make notes about your measurement, also, write down the course of the route.
- **Addition:** Every group can make one or more additional measurements, where they test unfavourable conditions on purpose, or comparability (by using known fixpoints). Think about such scenarios and discuss them with your exercise supervisor in advance.

Notes

wait for some seconds in a stationary point (80th seconds of the starting epoch)

wait under shadow (225th seconds of t0 epoch for approx. 10 seconds)

Tilt the receiver (290th seconds of t0 epoch for approx. 10 seconds)

4 Additional literature

- <https://emlid.com/reachrs2plus>
Homepage and internal structure of the Reach RS2 series
- <https://docs.emlid.com/reachrs2/>
Documentation Reach RS2, under Specification you can find:
- <http://files.emlid.com/docs/Datasheet%20RS2%20ENG%20web.pdf>
Datasheet
- <https://files.emlid.com/docs/reach-rs2-rs2plus-drawing.pdf>
Dimensions and annotations about the receivers